

MD. ZAWADUL ISLAM

Electrical & Electronic Engineering Undergraduate

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SUMMARY

Electrical and Electronic Engineering undergraduate with hands-on experience in power system protection, robotics, embedded systems, digital electronics, measurement and instrumentation, and signal processing. Experienced in Arduino-based control, MATLAB analysis, circuit design and simulation, and practical hardware implementation. Proven leadership in technical organizations and engineering events, with strong interest in power systems, protection, automation, and applied engineering design.

EDUCATION

Ahsanullah University of Science and Technology (AUST)

Sep 2022 – Dec 2026

BSc in Electrical & Electronic Engineering; CGPA: 3.636

Chittagong Cantonment Public College

Jun 2019 - Dec 2021

Higher Secondary Certificate; GPA: 5.00

Halishahar Cantonment Public College

2017 - 2019

Secondary School Certificate; GPA: 5.00

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Vice President, AUST Robotics Club

Jan 2026 – July 2026

- Lead technical and organizational operations of the club and coordinate robotics, embedded-system projects, workshops, and technical activities.
- Supervise technical teams for engineering events and competitions, coordinate inter-department collaboration, and support technical communication and participant engagement.

Organizing Director, AIAA AUST Student Branch

Dec 2024 - Jun 2025

- Planned and executed engineering seminars, workshops, and technical events; coordinated organizing, technical, design, and promotion teams.
- Managed participant communication, registrations, task distribution, timelines, and event logistics.

Campus Partner, Ulterior Engineering Ltd.

2024 - 2025

- Represented the organization within the university community and supported communication and engagement with students.
- Promoted engineering opportunities, initiatives, and student-focused activities on campus.

Brand Representative, Interactive Cares

2024 - 2025

- Represented Interactive Cares within the university community and promoted educational programs and student-focused opportunities.

SELECTED ENGINEERING PROJECTS

Overload Detection & Protection System - Arduino / Power Systems

- Built an ACS712-based current monitoring and protection system with a 700 mA trip threshold, dual-relay load isolation, buzzer indication, and I2C LCD monitoring.
- Validated normal and overload operation experimentally: 0.43 A measured for one 100 W bulb and 0.88 A for two bulbs, with successful protective tripping.

Obstacle Detecting & Object Handling Bot - Arduino

- Developed an autonomous mobile robot using Arduino Uno, HC-SR04 ultrasonic sensing, L298N motor control, and servo-based robotic arm/gripper operation.
- Led Arduino programming and contributed substantially to physical bot integration, presentation preparation, and demonstration.

AR Frequency Estimation of Voice Signal - MATLAB (DSP)

- Implemented Burg, Yule-Walker, and Levinson-Durbin autoregressive methods to model a recorded voice signal.
- Estimated the first seven dominant frequencies and compared frequency-estimation behavior and model performance.

Digital Voting System Using Sequential Logic Circuits

- Designed a real-time vote counting and display system using 74LS90 counters, 74LS47 BCD-to-seven-segment decoders, and 74HC85 comparator logic.
- Implemented debounced vote inputs, reset functionality, seven-segment tally display, and real-time highest-vote comparison.

AM Receiver Design & Implementation

- Applied communication-theory concepts to AM receiver hardware, tuning, noise-mitigation considerations, testing, and troubleshooting in an open-ended laboratory project.

SKILLS

Programming & Embedded: Arduino Programming, Embedded C / Microcontroller Logic, Sensor Interfacing, Motor & Servo Control

Engineering & Simulation: MATLAB (DSP, System Analysis), Proteus, DipTrace, Cadence Virtuoso, COMSOL, AutoCAD, Tinkercad

Electrical & Technical: Power Systems, Protection, Circuit Analysis & Design, Digital Electronics, Measurement & Instrumentation, Signal Processing

Professional: Technical Leadership, Project Coordination, Event Management, Team Collaboration, Technical Documentation, MS Office, MS Excel

ADDITIONAL INFORMATION

Languages: English, Bengali, Hindi, Urdu

Awards / Activities: RoboMania 2.0 - Lead; AUST Rover Challenge 2.0 - Segment Co-Lead; Robo Soccer, AUST CSE Carnival - 1st Runner-up

Relevant Coursework: Power System Protection, Power System II, Digital Signal Processing, Communication Theory, Microprocessor Unit Lab, Digital Electronics, Measurement & Instrumentation, Electrical Machines